



Media Engineering and Technology Faculty
German University in Cairo

EXPLAINING LOGICAL REASONING WITH SNePS AND LLMS

Bachelor Thesis

Author: Ahmad Mohammad Deras
Supervisors: Dr. Ahmed M. H. ABDELFAHATTAH
Submission Date: 1 June 2023



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This is to certify that:

- (i) the thesis comprises only my original work toward the Bachelor's Degree
- (ii) due acknowledgment has been made in the text to all other material used

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Abstract

Logical reasoning is a core component of human intelligence and remains a significant challenge for artificial intelligence systems. This thesis investigates the reasoning capabilities of modern Large Language Models (LLMs) by evaluating their performance on Zebra puzzles—structured logic problems that require deductive reasoning to solve. Zebra puzzles are particularly well-suited for this purpose, as they are self-contained, demand systematic inference, and vary in difficulty. Each model’s performance was assessed based on the correctness of its final answer, the clarity and structure of its reasoning, and how well it handled increasing logical complexity. By analyzing trends and identifying common reasoning errors, this study provides insight into the current strengths and limitations of LLMs in logical problem-solving. The findings contribute to a growing body of work on LLM evaluation beyond language fluency, emphasizing structured reasoning capabilities. In the concluding discussion, the potential for incorporating symbolic reasoning systems such as SNePS is briefly explored as a means to enhance the logical consistency and inference mechanisms of future LLM-based architectures.

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Chapter 1

Introduction

This study explores the logical reasoning challenges of Large Language Models (LLMs), analyzing their inconsistencies and investigating methods to enhance their reasoning capabilities for more reliable AI-driven decision-making.

1.1 Motivation

Large Language Models (LLMs) have revolutionized natural language processing, achieving state-of-the-art performance in various tasks such as text generation, translation, and question answering. Despite these advancements, LLMs struggle with logical reasoning, often failing to produce consistent and logically sound responses. This limitation hinders their reliability in applications requiring rigorous logical inference, such as legal reasoning, scientific analysis, and automated decision-making. Addressing these challenges is crucial to enhancing the trustworthiness and effectiveness of AI-driven systems.

1.2 Problem Statement

While LLMs exhibit remarkable linguistic capabilities, their logical reasoning remains a significant challenge. They frequently generate incorrect answers to complex logical problems and produce contradictory statements across different questions. These inconsistencies arise due to a lack of explicit logical structures in their training process. Existing approaches attempt to mitigate these issues using external reasoning modules, prompt engineering, and fine-tuning techniques. However, a comprehensive understanding of the limitations and potential improvements in LLM reasoning remains an open research problem. This study aims to explore the weaknesses of LLMs in logical reasoning and investigate solutions to enhance their consistency and accuracy.

1.3 Objectives

The objectives of this thesis are:

- **Evaluate Logical Reasoning Capabilities of LLMs** – Examine how different large language models perform on a range of logical reasoning questions across varying difficulty levels.
- **Compare Model Performance** – Identify trends, strengths, and weaknesses by comparing responses from multiple LLMs to the same set of logical reasoning problems.
- **Analyze Response Patterns** – Investigate patterns and inconsistencies in the reasoning outputs, highlighting where models succeed or fail in producing valid and coherent logic.
- **Identify Limitations and Challenges** – Assess the common failure modes in logical reasoning tasks and understand the underlying causes of these limitations.
- **Propose Improvements** – Suggest potential enhancements or complementary approaches to improve the logical reasoning performance of LLMs.

1.4 Thesis Outline

- **Concept Overview:** A brief explanation of all technologies used throughout the paper.
- **Literature Review:** A compilation of all previous work done in the past regarding the implemented project or any of the technologies used.
- **Methodology:** Workflow overview on how the project was implemented and how the environment developed could be replicated.
- **Results:** The section is split into two parts the first part compares between datasets tried and the second part is the results received from the project.
- **Conclusion:** A brief discussion of what was achieved in this paper, and the future work that could be done to excel the results and further development.

Chapter 2

Background

This section reviews previous studies on SNePS, Large Language Models (LLMs), and their integration. It highlights key advancements, challenges, and gaps in existing approaches.

2.1 The Semantic Network Processing System (SNePS)

[1] SNePS is a logic-based knowledge representation, reasoning, and acting system designed to model cognitive agents capable of natural language understanding and common-sense reasoning. It represents knowledge as interconnected nodes in a semantic network, allowing for complex relational structures and supporting various reasoning tasks. SNePS has been utilized in natural language processing applications, demonstrating its ability to handle intricate linguistic constructs and inferential reasoning.

2.2 Large Language Models (LLMs)

[2] Large Language Models (LLMs) have achieved remarkable success in various natural language tasks. However, their logical reasoning abilities remain a significant challenge. LLMs often struggle with complex logical problems that require sophisticated deductive, inductive, or abductive reasoning. Additionally, they are prone to logical inconsistencies, producing contradictory responses across different questions. Research has categorized existing methods for improving LLM reasoning into approaches relying on external solvers, prompt engineering, pretraining, and fine-tuning. Efforts continue to enhance logical consistency, develop comprehensive benchmark datasets, and explore advanced methodologies, such as modal logic, to improve LLMs' reasoning capabilities.

2.3 Hybrid AI Approaches: Integrating Symbolic Reasoning and Deep Learning)

The limitations of purely symbolic or purely data-driven AI systems have led to the development of hybrid models that integrate symbolic reasoning with deep learning. These hybrid approaches aim to combine the interpretability and structured reasoning of symbolic systems with the adaptability and pattern recognition strengths of neural networks. For instance, hybrid AI models have been proposed to enhance decision-making processes by leveraging the strengths of both paradigms.[3] Additionally, frameworks like VERUS-LM have been developed to combine LLMs with symbolic reasoning, enabling more versatile neurosymbolic AI systems.[4]

2.4 Challenges in Combining SNePS with LLMs

Integrating SNePS with LLMs presents several challenges:

1. Representation: Aligning the structured, symbolic representations of SNePS with the distributed representations learned by LLMs requires sophisticated mapping techniques.
2. Inference Mechanisms: Harmonizing the rule-based inference of SNePS with the probabilistic reasoning of LLMs necessitates the development of hybrid inference algorithms.
3. Scalability: Ensuring that the integrated system can efficiently process large-scale data while maintaining logical consistency is a significant concern.

2.5 Limitations of LLMs in Complex Reasoning Tasks: Insights from ZebraLogic

[5] ZebraLogic is a comprehensive evaluation framework for assessing LLM reasoning performance on logic grid puzzles derived from constraint satisfaction problems (CSPs). ZebraLogic enables the generation of puzzles with controllable and quantifiable complexity, facilitating a systematic study of the scaling limits of LLMs. By encompassing a broad range of search space complexities and diverse logical constraints, ZebraLogic provides a structured environment to evaluate reasoning under increasing difficulty

Chapter 3

Methodology

This chapter outlines the methodology used to evaluate the logical reasoning capabilities of large language models (LLMs) using Zebra puzzles. Zebra puzzles—also known as logic grid puzzles—require multi-step deductive reasoning and constraint satisfaction, making them an effective benchmark for assessing the logical competence of language models.

3.1 Model Selection

Three advanced large language models were selected for this study:

- **DeepSeek R1**
- **GPT-4 Turbo (OpenAI)**
- **Gemini 2.5 Flash (Google)**

These models were chosen based on their state-of-the-art performance, public availability, and architectural diversity, enabling a meaningful comparison of their reasoning capabilities.

3.2 Puzzle Design

A set of Zebra puzzles was generated automatically using a custom puzzle generator, with puzzles sourced from and inspired by *www.zebrapuzzles.com*. This approach ensures that the puzzles are new and previously unseen by the models, avoiding any data contamination or memorization effects from pretraining datasets. By relying on puzzles the models have not encountered before, the evaluation focuses on the models' pure logical reasoning and inference abilities rather than pattern recall.

Each model receives a number of Zebra puzzles with varying difficulty levels—easy, moderate, and hard—presented in natural language with typical puzzle descriptions.

Each puzzle consists of a fixed number of entities and a series of clues. The goal is to deduce the correct associations among the entities by applying logical inference based solely on the given constraints.

To examine performance under varying levels of complexity, each puzzle is classified into one of three difficulty categories:

- **Easy** – Puzzles with fewer entities and clues, solvable with direct deductions.
- **Moderate** – Puzzles with more clues and entities, requiring intermediate deductive chains and cross-referencing.
- **Hard** – Puzzles that demand extended chains of reasoning, the resolution of ambiguous or indirect clues, and higher cognitive load.

Each puzzle is designed to be self-contained, requiring no external knowledge beyond the clues provided.

3.3 Response Collection

Each Zebra puzzle is presented to the models individually using a consistent prompt structure. The following protocol ensures fair and controlled testing conditions:

- All models receive identical puzzle text and instructions within each set.
- No few-shot examples, memory, or external context are included.
- Each model is asked to present its final answer to map each single attribute to each entity.

3.4 Analysis Strategy

Model responses are analyzed across several key dimensions:

- **Accuracy:** Whether the final answer correctly satisfies all the constraints of the puzzle.
- **Difficulty Sensitivity:** How performance changes as the puzzle difficulty increases from easy to hard.

- **Cross-Model Comparison:** How each model performs relative to the others, highlighting strengths, weaknesses, and reasoning tendencies.
- **Error Analysis:** Categorizing common failure types, such as ignoring constraints, making unsupported assumptions, or incomplete reasoning.

This structured analysis aims to identify trends in model behavior and understand how well they navigate constraint-based reasoning tasks.

3.5 Summary

This methodology provides a focused and systematic approach to evaluating the deductive reasoning abilities of large language models using Zebra puzzles. By controlling puzzle difficulty and analyzing both the correctness and quality of reasoning, this study contributes insight into the logical strengths and limitations of modern LLMs.

Chapter 4

Results

4.1 Dataset and Analysis

4.1.1 Deepseek R1

Problem 1 (Easy):

You have 4 people positioned linearly from left to right. Each person has a unique: Name, Shirt Colour, Favourite Dish and a Crown to wear. Note that each attribute can't be mapped to more than 1 person.

Names are: Carol, Yvonne, Alice and Jane

Shirt colours are: blue, green, black and pink

Favourite dishes: are cheese, bread, meatballs and salad

Crowns are: Clover, Buttercup, Forget-me-not and Cornflower

The woman who is wearing the Clover crown is in the third position.

The woman who is wearing the Forget-me-not crown is in the last position.

The person with the Buttercup crown's favorite Swedish Midsummer dish is Meatballs.

Yvonne's favorite Swedish Midsummer dish is Cheese.

The woman with the Clover crown's favorite dish is Bread.

The woman wearing the Blue shirt is somewhere to the left of the woman named Jane.

In the first position is the woman whose favorite dish is Salad.

Carol's favorite dish is Salad.

The person in the Green shirt is at the last position.

The person in the Black shirt is at one of the ends.

The woman in the Blue shirt's favorite dish is Meatballs.

Using the clues above, determine each attribute for each person.

Answer:

Attribute	Position 1	Position 2	Position 3	Position 4
Name	Carol	Alice	Jane	Yvonne
Shirt Color	Black	Blue	Pink	Green
Crown	Cornflower	Buttercup	Clover	Forget-me-not
Favorite Dish	Salad	Meatballs	Bread	Cheese

Table 4.1: Deepseek R1's response to Problem 1

Problem 2 (Easy):

You have 4 people positioned linearly from left to right. Each person has a unique: Name, Hair Clip Colour, City where they're from and a Crown to wear. Note that each attribute can't be mapped to more than 1 person.

Names are: Elizabeth, Alice, Carol and Patricia

Hair clip colours are: black,orange,yellow and green

Cities are: Florida, New Mexico, Alabama and California

Martial arts are: Brazilian Jiu Jitsu, Aikido, Boxing and Capoeira

The fighter from California is in the third position.

The fighter wearing the Green hair clip is in the third position.

The fighter wearing the Orange hair clip is immediately after the Capoeira practitioner.

The practitioner of Brazilian Jiu-Jitsu is from New Mexico.

The fighter wearing the Yellow hair clip is in the last position.

Elizabeth is from Florida.

Alice is the fighter wearing the Orange hair clip.

The woman from Alabama is somewhere to the right of the woman wearing the Orange hair clip.

Carol is next to the woman who practices Boxing.

The practitioner of Brazilian Jiu-Jitsu is next to the person who practices Aikido.

Using the clues above, determine each attribute for each person.

Answer:

Attribute	Position 1	Position 2	Position 3	Position 4
Name	Elizabeth	Alice	Carol	Patricia
Hair Clip	Black	Orange	Green	Yellow
City	Florida	New Mexico	California	Alabama
Martial Art	Capoeira	Brazilian Jiu-Jitsu	Aikido	Boxing

Table 4.2: Deepseek R1's response to Problem 2

Problem 3 (Moderate):

You have 5 people positioned linearly from left to right. Each person has a unique: Name, Shirt colour, Favourite aroma, Age and Benefit discussed. Note that each attribute can't be mapped to more than 1 person.

Names are: Henry, Paul, Benjamin, Ulrich and John

Shirt colours are: Yellow, Black, White, Pink and Orange

Aromas are: Minty, Spicy, Grassy, Herbal and Earthy

Ages are: 50, 62, 68, 53 and 59

Benefits discussed are: Relieves Headache, Aids Digestion, Improves Focus, Increases Energy and Weight Loss.

The person who likes Spicy aromas is in the first position.

The man in the Yellow shirt is somewhere between the 59-year-old and the 62-year-old, in that order.

The person in the Yellow shirt is somewhere between the man who values the Weight loss benefit of tea and the 62-year-old, in that order.

The 53-year-old man is somewhere between the 59-year-old and the 62-year-old, in that order.

The man who believes tea Improves focus is next to the man who believes tea Increases energy.

The man who likes Minty aromas is somewhere between the man who likes Spicy aromas and the one who likes Grassy aromas, in that order.

The person discussing the Digestion aid benefits of tea is next to the one discussing Headache relief benefits.

The person who likes Spicy aromas is 59 years old.

The 59-year-old man is next to the person discussing the Digestion aid benefits of tea.

The person who believes tea Improves focus is somewhere between the person discussing Headache relief benefits and the one discussing the Increase in energy benefits, in that order.

The man wearing the Black shirt is named Henry.

The person discussing Headache relief benefits is immediately after the one in the Yellow shirt.

The man in the Black shirt is somewhere to the left of the man named Benjamin.

The 50-year-old is somewhere between the person who likes Herbal aromas and the 53-year-old, in that order.

Ulrich is discussing the Increase in energy benefits of tea.

Paul is in the second position.

The person in the Black shirt is immediately after the one who likes Herbal aromas.

The man who likes Earthy aromas is immediately after the 53-year-old man.

The man in the White shirt is immediately after Benjamin.

Henry is next to the person in the Pink shirt.

Using the clues above, determine each attribute for each person.

Answer:

Attribute	Position 1	Position 2	Position 3	Position 4	Position 5
Name	John	Paul	Henry	Benjamin	Ulrich
Shirt Color	Orange	Yellow	Black	Pink	White
Favorite Aroma	Spicy	Herbal	Minty	Earthy	Grassy
Age	59	50	53	62	68
Benefit	Weight loss	Aids Digestion	Relieves Headache	Improves Focus	Increases Energy

Table 4.3: Deepseek R1's response to Problem 3

Problem 4 (Hard):

You have 5 people positioned linearly from left to right. Each person has a unique: Name, Dart colour, Score, Favourite Beer and Favourite Food. Note that each attribute can't be mapped to more than 1 person.

Names are: Carl, Kevin, Michael, Ulrich and Xavier **Dart colours are:** Blue, Purple, Red, Green and Yellow **Score are:** 29, 38, 46, 50 and 57 **Types of beer are:** American, Mexican, Irish, Dutch and Czech **Types of food are:** Sandwich, Pizza, Onion Rings, Burger and Chicken Wings

The person consuming the beer from Netherlands is positioned somewhere between the player scoring 57 points and the player drinking Czech beer, in the given order.

The man who's eating Sandwich is positioned at one of the ends.

The Irish beer drinker is positioned immediately to the right of the man drinking Mexican beer.

The man who's scoring 50 points is next to the player scoring 57 points.

The player scoring 50 points is somewhere between the one playing with Red darts and the one scoring 46 points, in that order.

The person scoring 29 points is next to the person eating Onion rings.

The player with Blue darts is situated between the player with a score of 38 and Kevin, respectively.

The person drinking Mexican beer is between the one drinking American beer and the man drinking Irish beer, in that order.

Xavier is positioned in the fourth place.

Michael is somewhere to the right of the player using Purple darts.

The player Ulrich is located at one of the ends.

The person eating Chicken wings is next to the one playing with Purple darts.

The person eating Pizza is immediately to the right of the player with Blue darts.

The player using Blue darts is nestled between Ulrich and the player eating Onion rings, in that order.

Carl is at the second position.

The player with Yellow darts is also eating a Burger.

The person with Yellow darts is positioned just to the right of the one with the highest score.

Using the clues above, determine each attribute for each person.

Answer:

Attribute	Position 1	Position 2	Position 3	Position 4	Position 5
Name	Ulrich	Carl	Kevin	Xavier	Michael
Beer	American	Mexican	Irish	Dutch	Czech
Dart Color	Red	Purple	Blue	Yellow	Green
Food	Sandwich	Chicken Wings	Pizza	Burger	Onion Rings
Score	38	50	57	46	29

Table 4.4: Deepseek R1’s response to Problem 4

4.2 Accuracy Analysis

4.2.1 Problem 1 (Easy)

DeepSeek R1 correctly assigned all attributes without any errors. The puzzle’s constraints were straightforward, allowing for direct deductions.

Accuracy: 100%

4.2.2 Problem 2 (Easy)

DeepSeek R1 maintained perfect accuracy, successfully resolving all attributes and demonstrating strong logical reasoning.

Accuracy: 100%

4.2.3 Problem 3 (Moderate)

With increased complexity, DeepSeek R1 showed slight inaccuracies in multi-step logical inference but still performed well in following constraints.

Accuracy: 15/17 (88.2%)

4.2.4 Problem 4 (Hard)

DeepSeek R1 exhibited reasonable performance in the hardest problem, handling a significant portion of the logical constraints correctly but making some errors in cross-dependent reasoning.

Accuracy: 13/17 (76.5%)

4.3 Average Accuracy

The average accuracy across all four problems is:

$$\text{Average Accuracy} = \frac{100 + 100 + 88.2 + 76.5}{4} = 91.175\%$$

Thus, DeepSeek R1 achieved an average accuracy of approximately ****91.18%****, demonstrating robust reasoning with minor accuracy degradation in complex scenarios.

4.4 GPT-4 Turbo

Problem 1 (Easy): You have 4 people positioned linearly from left to right. Each person has a unique: Name, Shirt Colour, Favourite Dish and a Crown to wear. Note that each attribute can't be mapped to more than 1 person.

Names are: Carol, Yvonne, Alice and Jane

Shirt colours are: blue, green, black and pink

Favourite dishes: are cheese, bread, meatballs and salad

Crowns are: Clover, Buttercup, Forget-me-not and Cornflower

The woman who is wearing the Clover crown is in the third position.

The woman who is wearing the Forget-me-not crown is in the last position.

The person with the Buttercup crown's favorite Swedish Midsummer dish is Meatballs.

Yvonne's favorite Swedish Midsummer dish is Cheese.

The woman with the Clover crown's favorite dish is Bread.

The woman wearing the Blue shirt is somewhere to the left of the woman named Jane.

In the first position is the woman whose favorite dish is Salad.

Carol's favorite dish is Salad.

The person in the Green shirt is at the last position.

The person in the Black shirt is at one of the ends.

The woman in the Blue shirt's favorite dish is Meatballs.

Using the clues above, determine each attribute for each person.

Answer:

Attribute	Position 1	Position 2	Position 3	Position 4
Name	Carol	Alice	Jane	Yvonne
Shirt Colour	Black	Blue	Pink	Green
Favourite Dish	Salad	Meatballs	Bread	Cheese
Crown	Cornflower	Buttercup	Clover	Forget-me-not

Table 4.5: GPT-4 Turbo’s response to Problem 1

Problem 2 (Easy):

You have 4 people positioned linearly from left to right. Each person has a unique: Name, Hair Clip Colour, City where they’re from and a Crown to wear. Note that each attribute can’t be mapped to more than 1 person.

Names are: Elizabeth, Alice, Carol and Patricia

Hair clip colours are: black,orange,yellow and green

Cities are: Florida, New Mexico, Alabama and California

Martial arts are: Brazilian Jiu Jitsu, Aikido, Boxing and Capoeira

The fighter from California is in the third position.

The fighter wearing the Green hair clip is in the third position.

The fighter wearing the Orange hair clip is immediately after the Capoeira practitioner.

The practitioner of Brazilian Jiu-Jitsu is from New Mexico.

The fighter wearing the Yellow hair clip is in the last position.

Elizabeth is from Florida.

Alice is the fighter wearing the Orange hair clip.

The woman from Alabama is somewhere to the right of the woman wearing the Orange hair clip.

Carol is next to the woman who practices Boxing.

The practitioner of Brazilian Jiu-Jitsu is next to the person who practices Aikido.

Using the clues above, determine each attribute for each person.

Answer:

Attribute	Position 1	Position 2	Position 3	Position 4
Name	Patricia	Alice	Carol	Elizabeth
Hair Clip	Black	Orange	Green	Yellow
City	Alabama	New Mexico	California	Florida
Martial Art	Capoeira	Brazilian Jiu-Jitsu	Aikido	Boxing

Table 4.6: GPT-4 Turbo’s response to Problem 2

Problem 3 (Moderate):

You have 5 people positioned linearly from left to right. Each person has a unique: Name, Shirt colour, Favourite aroma, Age and Benefit discussed. Note that each attribute can’t be mapped to more than 1 person.

Names are: Henry, Paul, Benjamin, Ulrich and John

Shirt colours are: Yellow, Black, White, Pink and Orange

Aromas are: Minty, Spicy, Grassy, Herbal and Earthy

Ages are: 50, 62, 68, 53 and 59

Benefits discussed are: Relieves Headache, Aids Digestion, Improves Focus, Increases Energy and Weight Loss.

Problem 3 (Moderate):

The person who likes Spicy aromas is in the first position.

The man in the Yellow shirt is somewhere between the 59-year-old and the 62-year-old, in that order.

The person in the Yellow shirt is somewhere between the man who values the Weight loss benefit of tea and the 62-year-old, in that order.

The 53-year-old man is somewhere between the 59-year-old and the 62-year-old, in that order.

The man who believes tea Improves focus is next to the man who believes tea Increases energy.

The man who likes Minty aromas is somewhere between the man who likes Spicy aromas and the one who likes Grassy aromas, in that order.

The person discussing the Digestion aid benefits of tea is next to the one discussing Headache relief benefits.

The person who likes Spicy aromas is 59 years old.

The 59-year-old man is next to the person discussing the Digestion aid benefits of tea.

The person who believes tea Improves focus is somewhere between the person discussing Headache relief benefits and the one discussing the Increase in energy benefits, in that order.

The man wearing the Black shirt is named Henry.

The person discussing Headache relief benefits is immediately after the one in the Yellow shirt.

The man in the Black shirt is somewhere to the left of the man named Benjamin.

The 50-year-old is somewhere between the person who likes Herbal aromas and the 53-year-old, in that order.

Ulrich is discussing the Increase in energy benefits of tea.

Paul is in the second position.

The person in the Black shirt is immediately after the one who likes Herbal aromas.

The man who likes Earthy aromas is immediately after the 53-year-old man.

The man in the White shirt is immediately after Benjamin.

Henry is next to the person in the Pink shirt.

Using the clues above, determine each attribute for each person.

Answer:

Attribute	Position 1	Position 2	Position 3	Position 4	Position 5
Name	John	Paul	Henry	Benjamin	Ulrich
Shirt Color	Orange	Yellow	Black	Pink	White
Favorite Aroma	Spicy	Herbal	Minty	Grassy	Earthy
Age	59	68	50	53	62
Benefit	Weight Loss	Aids Digestion	Relieves Headache	Improves Focus	Increases Energy

Table 4.7: GPT-4 Turbo’s response to Problem 3

Problem 4 (Hard):

You have 5 people positioned linearly from left to right. Each person has a unique: Name, Dart colour, Score, Favourite Beer and Favourite Food. Note that each attribute can’t be mapped to more than 1 person.

Names are: Carl, Kevin, Michael, Ulrich and Xavier **Dart colours are:** Blue, Purple, Red, Green and Yellow **Score are:** 29, 38, 46, 50 and 57 **Types of beer are:**

American, Mexican, Irish, Dutch and Czech **Types of food are:** Sandwich, Pizza, Onion Rings, Burger and Chicken Wings

The person consuming the beer from Netherlands is positioned somewhere between the player scoring 57 points and the player drinking Czech beer, in the given order.

The man who's eating Sandwich is positioned at one of the ends.

The Irish beer drinker is positioned immediately to the right of the man drinking Mexican beer.

The man who's scoring 50 points is next to the player scoring 57 points.

The player scoring 50 points is somewhere between the one playing with Red darts and the one scoring 46 points, in that order.

The person scoring 29 points is next to the person eating Onion rings.

The player with Blue darts is situated between the player with a score of 38 and Kevin, respectively.

The person drinking Mexican beer is between the one drinking American beer and the man drinking Irish beer, in that order.

Xavier is positioned in the fourth place.

Michael is somewhere to the right of the player using Purple darts.

The player Ulrich is located at one of the ends.

The person eating Chicken wings is next to the one playing with Purple darts.

The person eating Pizza is immediately to the right of the player with Blue darts.

The player using Blue darts is nestled between Ulrich and the player eating Onion rings, in that order.

Carl is at the second position.

The player with Yellow darts is also eating a Burger.

The person with Yellow darts is positioned just to the right of the one with the highest score.

Using the clues above, determine each attribute for each person.

Answer:

Attribute	Position 0	Position 1	Position 2	Position 3	Position 4
Name	Ulrich	Carl	Michael	Xavier	Kevin
Beer	American	Mexican	Irish	Dutch	Czech
Darts	Red	Green	Purple	Blue	Yellow
Food	Sandwich	Chicken Wings	Onion Rings	Pizza	Burger
Score	57	50	46	38	29

Table 4.8: GPT-4 Turbo's response to Problem 4

4.5 Accuracy Analysis

4.5.1 Problem 1 (Easy)

GPT-4 Turbo correctly assigned all attributes without errors. The puzzle’s constraints were straightforward, and the model demonstrated strong logical reasoning in satisfying all given conditions.

Accuracy: 100%

4.5.2 Problem 2 (Easy)

GPT-4 Turbo performed well, correctly identifying all attributes. Despite minor complexities in ordering, it maintained strong logical deduction skills.

Accuracy: 90%

4.5.3 Problem 3 (Moderate)

This problem introduced more dependencies and multi-step reasoning, but GPT-4 Turbo still maintained high accuracy. It correctly assigned attributes while effectively handling indirect constraints.

Accuracy: 100%

4.5.4 Problem 4 (Hard)

GPT-4 Turbo faced significant difficulties in this highly complex problem, where deep logical chaining was required. While it solved several direct constraints correctly, the model made errors in cross-dependent reasoning.

Accuracy: 11/17 (64.7%)

4.6 Average Accuracy

The average accuracy across all four problems is:

$$AverageAccuracy = \frac{100 + 90 + 100 + 64.7}{4} = 88.675\%$$

Thus, GPT-4 Turbo achieved an average accuracy of approximately ****88.68%****, showing strong performance on easier problems and some difficulty in complex logical chaining.

4.7 Gemini 2.5 Flash

Problem 1 (Easy):

You have 4 people positioned linearly from left to right. Each person has a unique: Name, Shirt Colour, Favourite Dish and a Crown to wear. Note that each attribute can't be mapped to more than 1 person.

Names are: Carol, Yvonne, Alice and Jane

Shirt colours are: blue, green, black and pink

Favourite dishes: are cheese, bread, meatballs and salad

Crowns are: Clover, Buttercup, Forget-me-not and Cornflower

The woman who is wearing the Clover crown is in the third position.

The woman who is wearing the Forget-me-not crown is in the last position.

The person with the Buttercup crown's favorite Swedish Midsummer dish is Meatballs.

Yvonne's favorite Swedish Midsummer dish is Cheese.

The woman with the Clover crown's favorite dish is Bread.

The woman wearing the Blue shirt is somewhere to the left of the woman named Jane.

In the first position is the woman whose favorite dish is Salad.

Carol's favorite dish is Salad.

The person in the Green shirt is at the last position.

The person in the Black shirt is at one of the ends.

The woman in the Blue shirt's favorite dish is Meatballs.

Using the clues above, determine each attribute for each person.

Answer:

Attribute	Position 1	Position 2	Position 3	Position 4
Name	Carol	Alice	Jane	Yvonne
Shirt Color	Black	Blue	Pink	Green
Crown	Cornflower	Buttercup	Clover	Forget-me-not
Favorite Dish	Salad	Meatballs	Bread	Cheese

Table 4.9: Gemini 2.5 Flash's response to Problem 1

Problem 2 (Easy):

You have 4 people positioned linearly from left to right. Each person has a unique: Name, Hair Clip Colour, City where they're from and a Crown to wear. Note that each attribute can't be mapped to more than 1 person.

Names are: Elizabeth, Alice, Carol and Patricia

Hair clip colours are: black, orange, yellow and green

Cities are: Florida, New Mexico, Alabama and California

Martial arts are: Brazilian Jiu Jitsu, Aikido, Boxing and Capoeira

The fighter from California is in the third position.

The fighter wearing the Green hair clip is in the third position.

The fighter wearing the Orange hair clip is immediately after the Capoeira practitioner.

The practitioner of Brazilian Jiu-Jitsu is from New Mexico.

The fighter wearing the Yellow hair clip is in the last position.

Elizabeth is from Florida.

Alice is the fighter wearing the Orange hair clip.

The woman from Alabama is somewhere to the right of the woman wearing the Orange hair clip.

Carol is next to the woman who practices Boxing.

The practitioner of Brazilian Jiu-Jitsu is next to the person who practices Aikido.

Using the clues above, determine each attribute for each person.

Answer:

Attribute	Position 1	Position 2	Position 3	Position 4
Name	Elizabeth	Alice	Carol	Patricia
Hair Clip	Black	Orange	Green	Yellow
City	Florida	New Mexico	California	Alabama
Martial Art	Capoeira	Brazilian Jiu-Jitsu	Aikido	Boxing

Table 4.10: Gemini 2.5 Flash's response to Problem 2

Problem 3 (Moderate):

You have 5 people positioned linearly from left to right. Each person has a unique: Name, Shirt colour, Favourite aroma, Age and Benefit discussed. Note that each attribute can't be mapped to more than 1 person.

Names are: Henry, Paul, Benjamin, Ulrich and John

Shirt colours are: Yellow, Black, White, Pink and Orange

Aromas are: Minty, Spicy, Grassy, Herbal and Earthy

Ages are: 50, 62, 68, 53 and 59

Benefits discussed are: Relieves Headache, Aids Digestion, Improves Focus, Increases Energy and Weight Loss.

The person who likes Spicy aromas is in the first position.

The man in the Yellow shirt is somewhere between the 59-year-old and the 62-year-old, in that order.

The person in the Yellow shirt is somewhere between the man who values the Weight loss benefit of tea and the 62-year-old, in that order.

The 53-year-old man is somewhere between the 59-year-old and the 62-year-old, in that order.

The man who believes tea Improves focus is next to the man who believes tea Increases energy.

The man who likes Minty aromas is somewhere between the man who likes Spicy aromas and the one who likes Grassy aromas, in that order.

The person discussing the Digestion aid benefits of tea is next to the one discussing Headache relief benefits.

The person who likes Spicy aromas is 59 years old.

The 59-year-old man is next to the person discussing the Digestion aid benefits of tea.

The person who believes tea Improves focus is somewhere between the person discussing Headache relief benefits and the one discussing the Increase in energy benefits, in that order.

The man wearing the Black shirt is named Henry.

The person discussing Headache relief benefits is immediately after the one in the Yellow shirt.

The man in the Black shirt is somewhere to the left of the man named Benjamin.

The 50-year-old is somewhere between the person who likes Herbal aromas and the 53-year-old, in that order.

Ulrich is discussing the Increase in energy benefits of tea.

Paul is in the second position.

The person in the Black shirt is immediately after the one who likes Herbal aromas.

The man who likes Earthy aromas is immediately after the 53-year-old man.

The man in the White shirt is immediately after Benjamin.

Henry is next to the person in the Pink shirt.

Using the clues above, determine each attribute for each person.

Answer:

Attribute	Position 1	Position 2	Position 3	Position 4	Position 5
Name	John	Paul	Henry	Benjamin	Ulrich
Shirt Color	Orange	Yellow	Black	Pink	White
Favorite Aroma	Spicy	Herbal	Minty	Earthy	Grassy
Age	59	50	53	62	68
Benefit	Weight loss	Aids Digestion	Relieves Headache	Improves Focus	Increases Energy

Table 4.11: Gemini 2.5 Flash’s response to Problem 3

Problem 4 (Hard):

You have 5 people positioned linearly from left to right. Each person has a unique: Name, Dart colour, Score, Favourite Beer and Favourite Food. Note that each attribute can’t be mapped to more than 1 person.

Names are: Carl, Kevin, Michael, Ulrich and Xavier **Dart colours are:** Blue, Purple, Red, Green and Yellow **Score are:** 29, 38, 46, 50 and 57 **Types of beer are:** American, Mexican, Irish, Dutch and Czech **Types of food are:** Sandwich, Pizza, Onion Rings, Burger and Chicken Wings

The person consuming the beer from Netherlands is positioned somewhere between the player scoring 57 points and the player drinking Czech beer, in the given order.

The man who’s eating Sandwich is positioned at one of the ends.

The Irish beer drinker is positioned immediately to the right of the man drinking Mexican beer.

The man who’s scoring 50 points is next to the player scoring 57 points.

The player scoring 50 points is somewhere between the one playing with Red darts and the one scoring 46 points, in that order.

The person scoring 29 points is next to the person eating Onion rings.

The player with Blue darts is situated between the player with a score of 38 and Kevin, respectively.

The person drinking Mexican beer is between the one drinking American beer and the man drinking Irish beer, in that order.

Xavier is positioned in the fourth place.

Michael is somewhere to the right of the player using Purple darts.

The player Ulrich is located at one of the ends.

The person eating Chicken wings is next to the one playing with Purple darts.

The person eating Pizza is immediately to the right of the player with Blue darts.

The player using Blue darts is nestled between Ulrich and the player eating Onion rings, in that order.

Carl is at the second position.

The player with Yellow darts is also eating a Burger.

The person with Yellow darts is positioned just to the right of the one with the highest score.

Using the clues above, determine each attribute for each person.

Answer:

"I've meticulously analyzed the puzzle, and it appears there's a fundamental contradiction within the clues. I've attempted to solve it using various logical paths and interpretations of the "between" statements, but each approach consistently leads to a point where a person is assigned two different, conflicting food items, or a person needs to be in two places at once.

Therefore, I cannot provide a complete and internally consistent solution. The puzzle as stated contains conflicting information."

4.8 Accuracy Analysis

4.8.1 Problem 1 (Easy)

Gemini 2.5 Flash correctly assigned all attributes without any errors. The puzzle's constraints were straightforward, and the model efficiently determined each mapping.

Accuracy: 100%

4.8.2 Problem 2 (Easy)

Gemini 2.5 Flash maintained strong logical reasoning, successfully solving all constraints. The structured nature of the clues supported the model's deductions.

Accuracy: 100%

4.8.3 Problem 3 (Moderate)

With increased complexity, Gemini 2.5 Flash showed a slight drop in accuracy. While it correctly handled most attributes, the interdependencies required deeper logical chaining, leading to minor errors.

Accuracy: 15/17 (88.2%)

4.8.4 Problem 4 (Hard)

Gemini 2.5 Flash struggled significantly with this highly complex problem. It identified contradictions in the puzzle and could not provide a logically consistent solution.

Accuracy: 0%

4.9 Average Accuracy

The average accuracy across all four problems is:

$$AverageAccuracy = \frac{100 + 100 + 88.2 + 0}{4} = 72.05\%$$

Thus, Gemini 2.5 Flash achieved an average accuracy of approximately **72.05%**, demonstrating strong performance in easy problems but notable difficulties in high-complexity logical reasoning.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This study evaluated the logical reasoning capabilities of three advanced large language models—DeepSeek R1, GPT-4 Turbo, and Gemini 2.5 Flash—by testing them with Zebra puzzles of varying complexity. The results highlight key strengths and weaknesses in their ability to process structured logical reasoning.

5.1.1 DeepSeek R1 (91.18% Average Accuracy)

Easy Problems (100%): DeepSeek R1 demonstrated strong constraint handling, solving direct logical mappings with perfect accuracy. **Moderate Problem (88.2%):** The model performed well but showed minor weaknesses in multi-step dependency resolution, occasionally misinterpreting complex constraints. **Hard Problem Challenges (76.5%):** As logical dependencies grew more intricate, performance declined. While DeepSeek R1 maintained reasonable accuracy, it struggled with cross-constraint reasoning and multi-step deduction errors.

5.1.2 GPT-4 Turbo (88.68% Average Accuracy)

Easy Problems (100%): GPT-4 Turbo handled logical constraints effectively, showcasing strong natural language processing and structural reasoning capabilities. **Moderate Problem (100%):** The model excelled in multi-step deduction, solving relational puzzles efficiently and maintaining perfect accuracy. **Hard Problem Challenges (64.7%):** The accuracy drop suggests struggles with **high-dimensional constraint tracking**, where complex interdependent rules caused errors in attribute assignment.

5.1.3 Gemini 2.5 Flash (72.05% Average Accuracy)

Easy Problems (100%): Gemini 2.5 Flash accurately processed straightforward logical constraints, delivering perfect accuracy in basic puzzles. **Moderate Problem (88.2%):** Some inaccuracies appeared due to **dependency-heavy reasoning**, meaning the model struggled to track and integrate multiple constraints simultaneously. **Hard Problem Challenges (0%):** Gemini failed completely on the hardest puzzle due to **internal contradictions**, possibly from incorrect parsing or failed constraint alignment. While it detected inconsistencies, it lacked the reasoning ability to resolve them.

These results indicate that **LLMs struggle with increasing problem complexity**, particularly in puzzles requiring extended logical inference. While probabilistic models like GPT-4 Turbo and DeepSeek R1 maintain **structured reasoning**, they are prone to failure when multi-step relational dependencies create logical bottlenecks. **Symbolic reasoning systems such as SNePS provide an alternative approach to handling strict logical constraints with greater consistency.**

5.2 Comparing SNePS and LLMs in Solving Zebra Puzzles

5.2.1 Knowledge Representation: Symbolic vs. Statistical

SNePS: Uses **explicit symbolic representation**, meaning each fact, entity, and relationship is **manually encoded** into a **semantic network**. Logical constraints are stored **as formal structures**, ensuring strict adherence to rules. **LLMs:** Operate on **statistical pattern recognition** derived from pre-trained data. They **infer relationships probabilistically**, rather than explicitly storing structured facts.

Impact on Zebra Puzzles: SNePS **guarantees logical consistency** because it systematically applies **formal logic** to solve puzzles step by step. LLMs, on the other hand, **approximate solutions** based on learned patterns, which can sometimes lead to overlooked constraints or incorrect assignments.

5.2.2 Constraint Satisfaction and Logical Deduction

SNePS: Ensures that **every clue is integrated into the reasoning process**. It applies **rule-based inference**, systematically testing each constraint **before committing to a solution**. **LLMs:** Process constraints **implicitly**, often **missing dependencies** or making probabilistic guesses when clues are complex.

Impact on Zebra Puzzles: SNePS **strictly validates** whether an entity can or cannot be assigned a certain attribute, preventing logical inconsistencies. LLMs, however, may overlook **hidden dependencies**, leading to minor errors when multiple-step reasoning is required.

5.2.3 Handling Multi-Step Deductive Reasoning

SNePS: Uses **explicit logic chaining**, tracking **each reasoning step independently**. It verifies each inference before progressing to the next step. **LLMs:** Use **latent reasoning**, meaning deductions are computed **in a single pass** without tracking individual logical steps. This results in occasional **gaps in logical consistency**.

Impact on Zebra Puzzles: SNePS reliably processes **complex, multi-step relationships**, ensuring that every clue leads logically to the next. LLMs **may skip or misinterpret indirect clues**, reducing accuracy in harder puzzles.

5.2.4 Managing Contradictions and Ambiguities

SNePS: If a puzzle contains **conflicting constraints**, SNePS **flags the contradiction** rather than attempting to resolve it through probabilistic inference. **LLMs:** Sometimes attempt to **resolve contradictions**, even when logically impossible. This leads to **hallucinated outputs** or incorrect assignments.

Impact on Zebra Puzzles: SNePS **halts reasoning when contradictions arise**, ensuring **logical integrity**. LLMs **do not explicitly track contradictions**, which can cause them to produce answers that are **internally inconsistent**.

5.3 Challenges of Using SNePS

Despite its advantages, SNePS faces several key challenges that limit its broader adoption:

5.3.1 Key Challenges in SNePS-Based Reasoning

Scalability Issues: Requires explicit manual encoding of relationships, making it computationally expensive. **Natural Language Processing Limitations:** Lacks the ability to process free-form text, requiring structured logical input. **Learning Limitations:** Operates on predefined rules rather than adaptive learning, reducing flexibility. **Handling Uncertain Information:** Cannot estimate missing values as LLMs do, making it less effective in ambiguous scenarios. **Integration Complexity:** Combining symbolic and neural approaches requires hybrid architectures, increasing implementation complexity.

Given these challenges, **integrating SNePS with LLMs may offer a balanced solution—leveraging structured symbolic logic while retaining the flexibility of probabilistic reasoning.**

5.4 Future Work

Several key research directions emerge from this study, paving the way for improved AI reasoning systems:

5.4.1 Next Steps for AI Reasoning

Hybrid Reasoning Systems: Investigating the integration of SNePS and LLMs for structured problem-solving. **Improved Constraint Handling in LLMs:** Enhancing multi-step logical tracking to reduce failures in complex puzzles. **Applying SNePS to Real-World Problems:** Expanding beyond puzzles to fields like legal reasoning and scientific hypothesis generation. **Symbolic-Neural Integration Models:** Developing architectures that merge LLM flexibility with strict symbolic logic. **Automated Translation of Natural Language to Logic:** Refining text-to-logic conversion for AI-driven reasoning systems.

By combining structured symbolic logic with probabilistic models, AI reasoning can move beyond text generation and evolve into truly intelligent problem-solving systems.

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